

2. (a) The following table gives information about some atoms, **A-E**.

A-E are not the chemical symbols of the elements.

Atom	Number of protons	Number of neutrons
A	12	12
B	10	10
C	6	8
D	11	12
E	6	6

(i) Give the letter, **A-E**, of the atom with a mass number of 12. [1]

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(ii) Give the letter, **A-E**, of the atom found in Group 1 of the Periodic Table. [1]

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(iii) Choose the letters, **A-E**, of the atoms that are isotopes. Give the reason for your choice. [2]

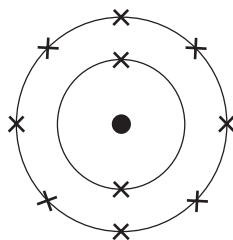
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(b) The diagram below shows the electronic structure of an element in the Periodic Table.



(i) Using $\times$ to represent an electron, draw a diagram to show the electronic structure of the element which lies directly **below** this one in the Periodic Table. [1]

(ii) Explain why both of these elements are chemically unreactive. [1]

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2. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Electrical conductor
oxygen	−219	−183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

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(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

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(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

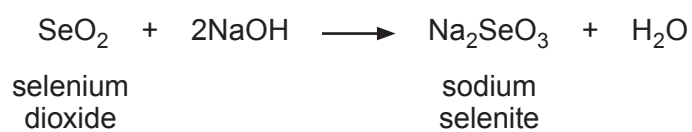
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- (b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



- (i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3 . [1]

$$A_r(\text{Na}) = 23 \quad A_r(\text{Se}) = 79 \quad A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

$$\text{Percentage} = \dots\dots\dots \%$$



2. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

- (i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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- (ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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- (b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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- (c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



2. (a) In 1915 Alfred Wegener suggested that the Earth’s continents were once joined together as one large land mass.



- (i) State **three** pieces of evidence that Alfred Wegener used to support his theory. [3]

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- (ii) Explain why Wegener’s theory was not originally accepted by other scientists, but it is today. [2]

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(b) What type of destructive event is likely to happen at a **conservative** plate boundary? [1]

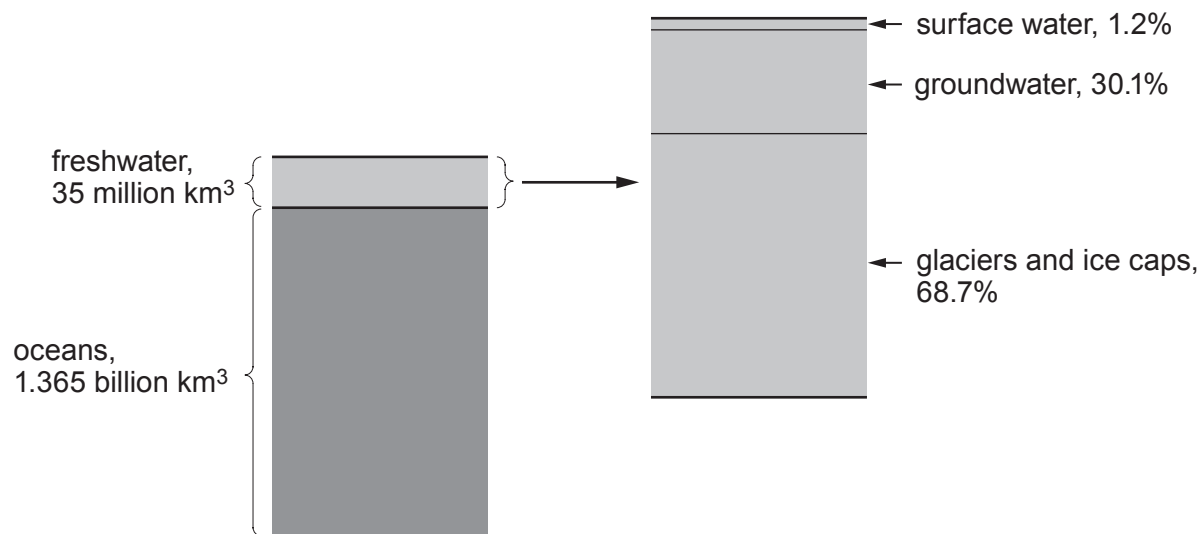
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2. The diagram shows the distribution of water on Earth.



(a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = million km³

(b) As the Earth’s population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination. [1]

Give a method that could be used to obtain freshwater from seawater.

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(c) Calcium ions, Ca^{2+} , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

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(d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

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